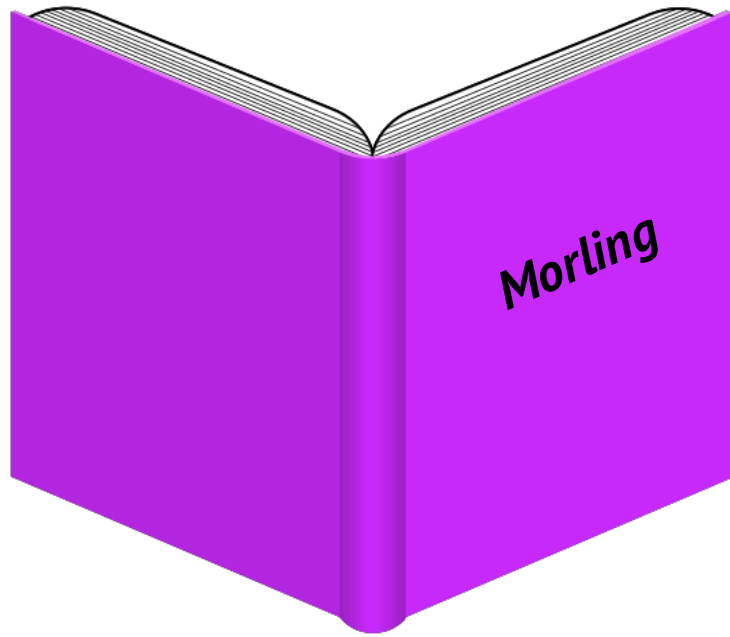




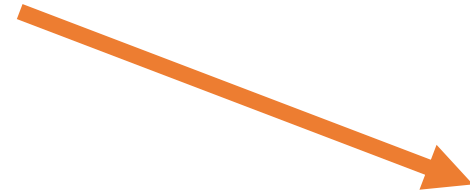
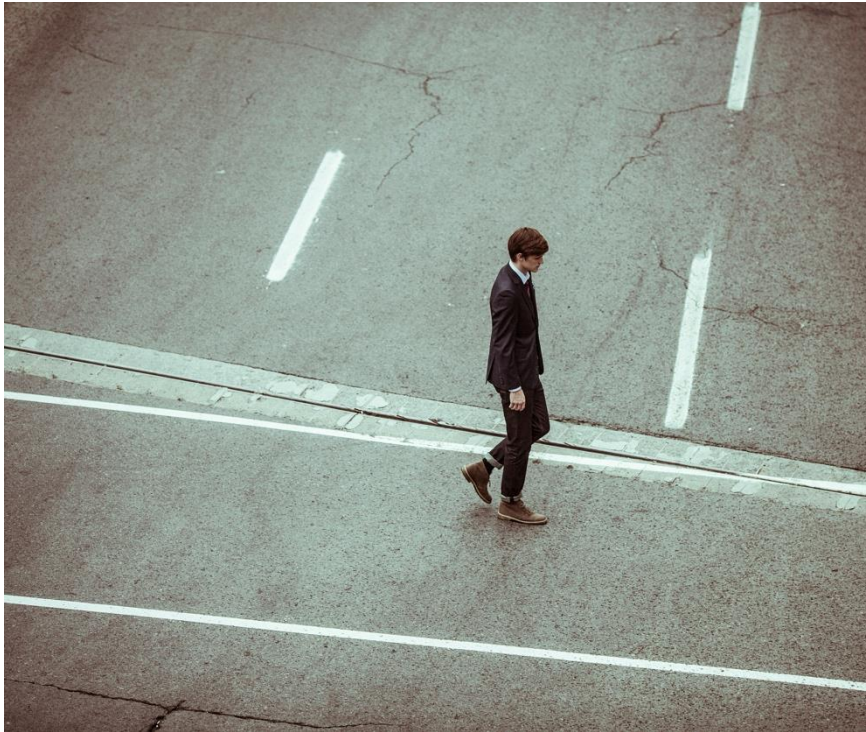
Research Methods and Statistics

Lecture 8: Experimental Research

Johnny van Doorn



Pictures source: pixabay



Do we take hot showers when we feel lonely?

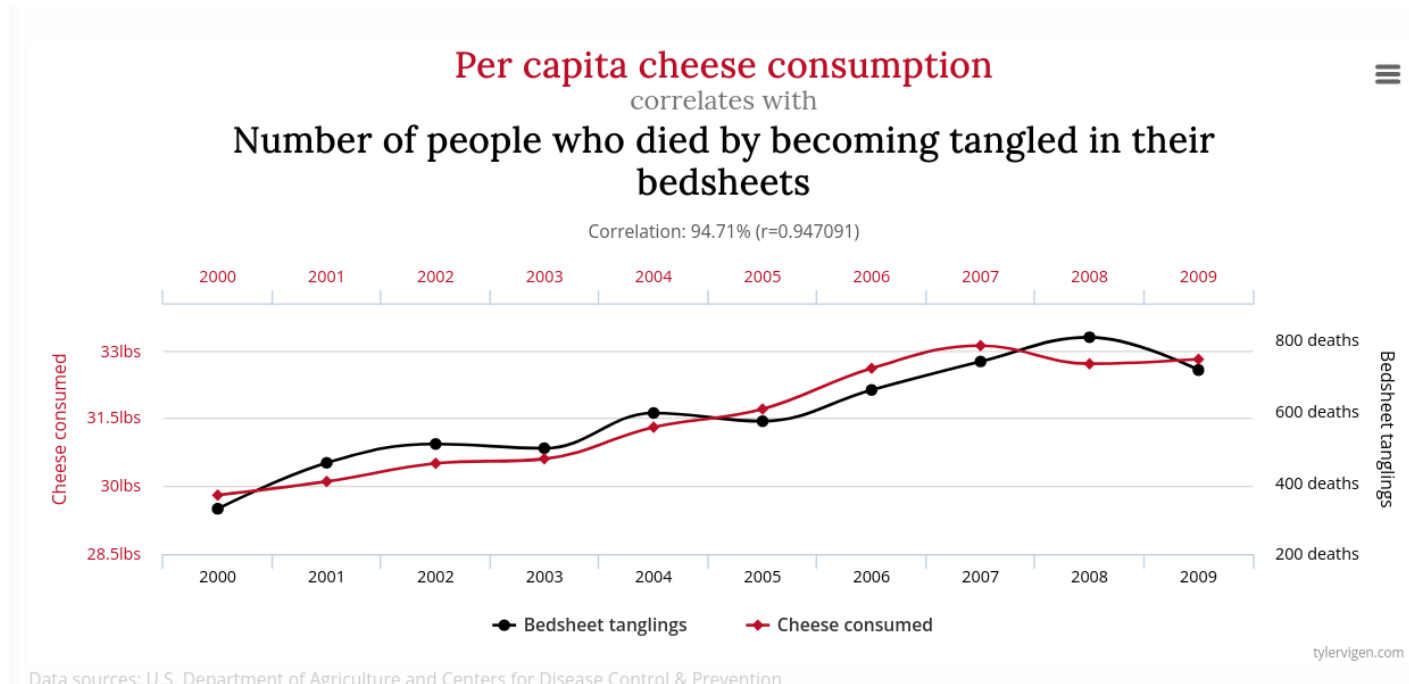


How can we make proper, causal claims?



Your Research Questions

- Think of two (psychological) variables you would like to investigate
- Need inspiration? → <http://www.tylervigen.com/spurious-correlations>



Today

1. Experimental studies vs observational studies

2. Pitfalls in experimental studies

1. Pseudoreplication

3. Different experimental designs

1. Pretest

2. Within groups vs between groups

4. Order effects

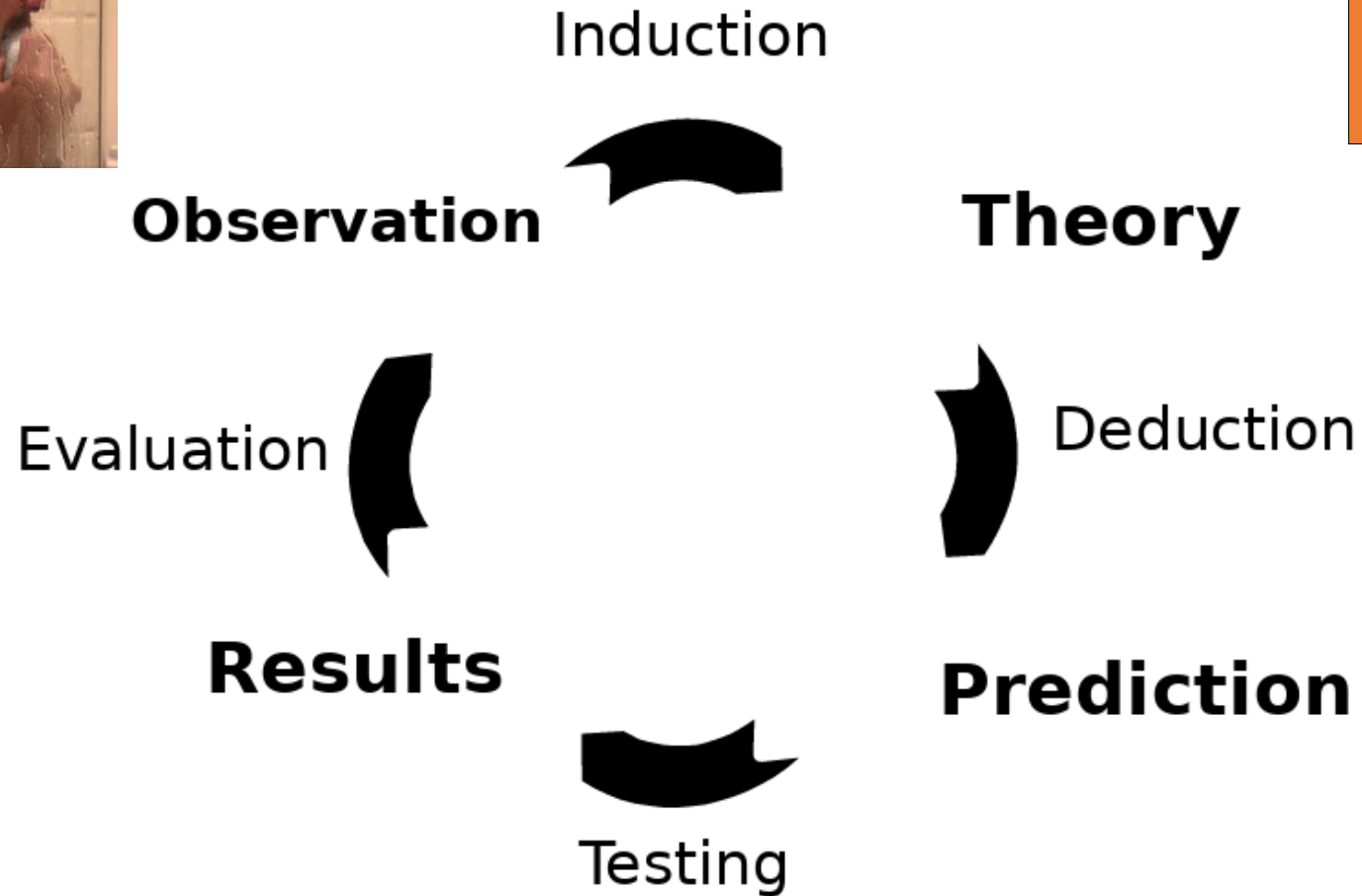
1. Counterbalancing

5. Recap

1. Next time

2. Example exam question

The empirical cycle



Theory: People use warm showers and baths to compensate for a lack of social warmth.

Hypothesis: Increased loneliness causes increased shower temperature



Parsimonious?
Falsifiable?

Observational study?

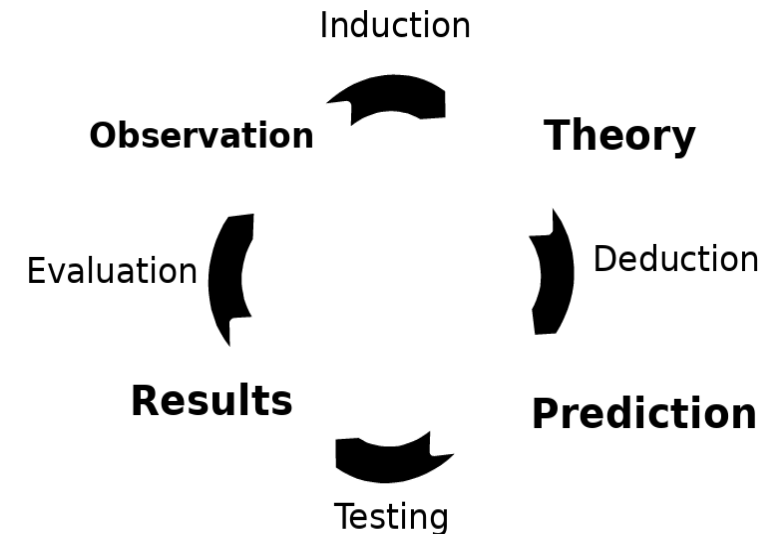
- What kind of observational study could you do?
- Bargh and Shalev (2012) hypothesized that people use warm showers and baths to compensate for a lack of social warmth

“Therefore, it seems that the “coldness” of loneliness or rejection could be treated somewhat successfully through the application of physical warmth—that is, physical and social warmth might be substitutable for each other at some level”

Hypothesis, Predictions, Operationalization

(in Bargh & Shalev)

- Hypothesis: Increased loneliness *causes* increased shower temperature
- Prediction: There is an *association* between loneliness and shower temperature
- Operationalization:
 - Dependent variable: Shower temperature
 - Independent variable: Loneliness



Operationalization of Shower Temperature

- *“How often do you usually take a bath?”* (on an 8-point scale ranging from more than 3 times a day to less than once a week)
- *“What temperature water do you use?”* (on a 6-point scale ranging from cold to very hot)
- *“About how much time do you spend in the bath?”* (on a 7-point scale ranging from less than 2 minutes to over 30 minutes)

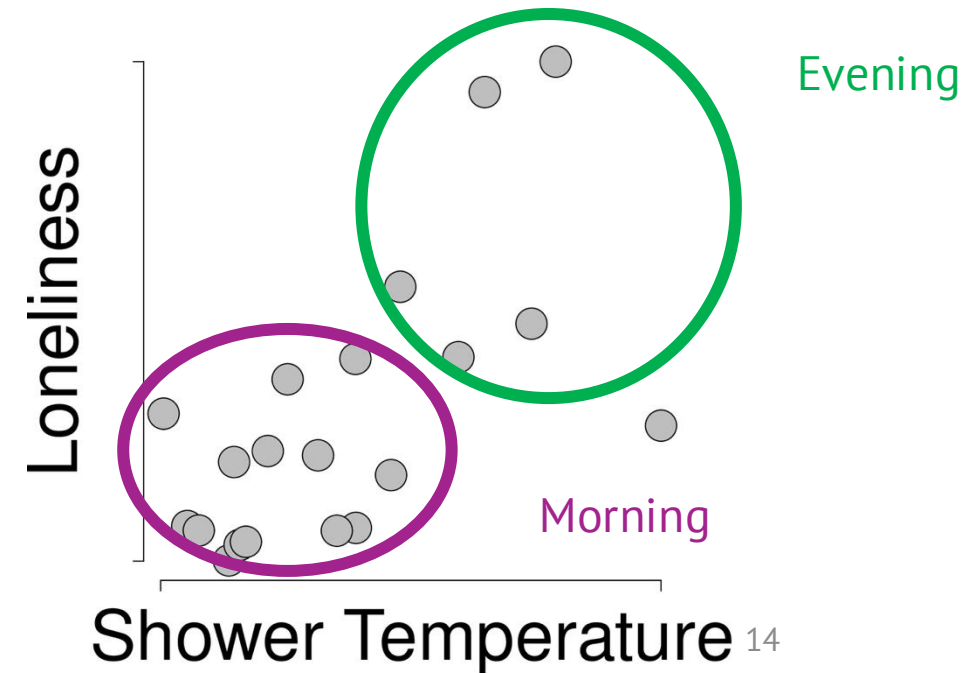
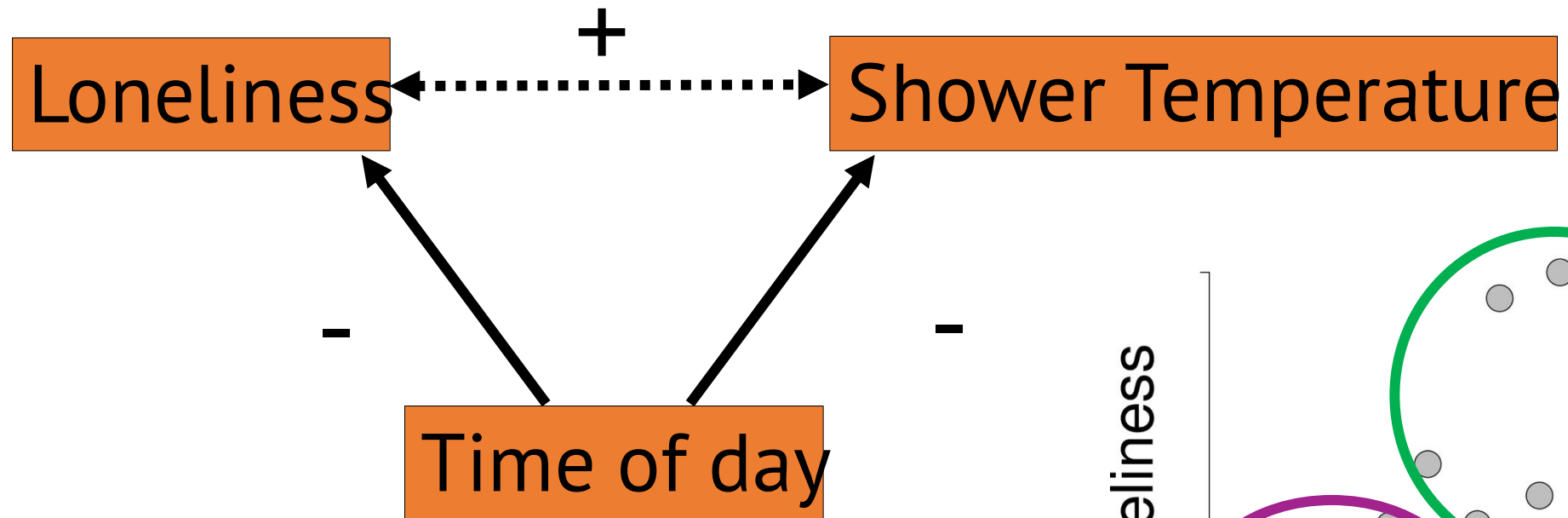
Operationalization of Loneliness

- *“I feel isolated from others”* (on a 4-point scale ranging from “never” to “often”)
- *“There is no one I can turn to”*
- *“My interests and ideas are not shared by those around me”*

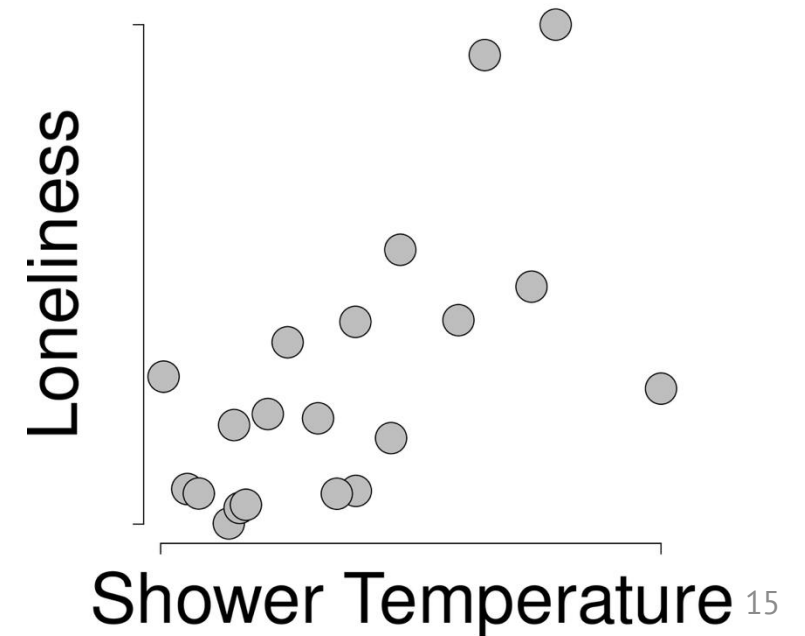
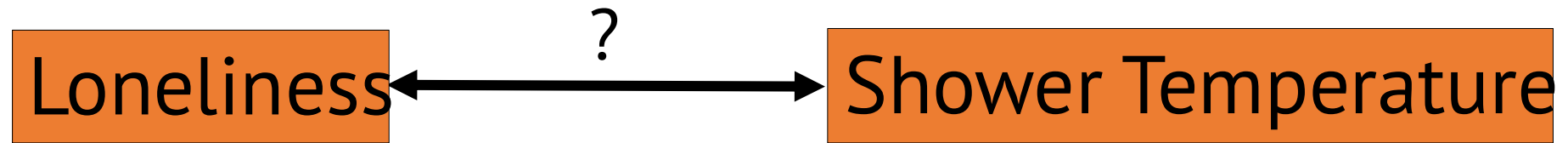
Russell, D., Peplau, L.A., & Cutrona, C.E. (1980). The revised UCLA Loneliness Scale: Concurrent and discriminant validity evidence. *Journal of Personality and Social Psychology*, 39, 472-480.

Take it yourself! <https://artemciy.gitlab.io/loneliness-scale/#/>

Confounding factors?



Directionality?



Three criteria for causal claims

(Morling, p. 283)

- Observational study (Bargh & Shalev, 2012):
 - ✓ **Covariance**: lonelier people take warmer showers
 - ✗ **Temporal precedence**: which came first?
 - ✗ **Internal validity**: third variables, e.g., time of day
- Experiments can establish all three

Experimental study

- What kind of experimental study could you do?

Hypothesis, Predictions, Operationalization

- Hypothesis: Increased loneliness *causes* taking warmer showers
- Prediction: *IF* loneliness increases, *THEN* the shower temperature increases
- Operationalization:
 - Dependent variable: Shower temperature
 - Independent variable: *Two groups of participants, with different levels of loneliness*

Experimental Manipulation of Loneliness

- Participants take questionnaire about extraversion/introversion, then get feedback:

Lonely group

“... people who score high on extraversion have trouble keeping their relationships stable later in life.”

Non-lonely group

“... being an extravert is a really good thing for relationships.”

Experimental Manipulation of Loneliness

- Participants take questionnaire about extraversion/introversion, then get feedback:

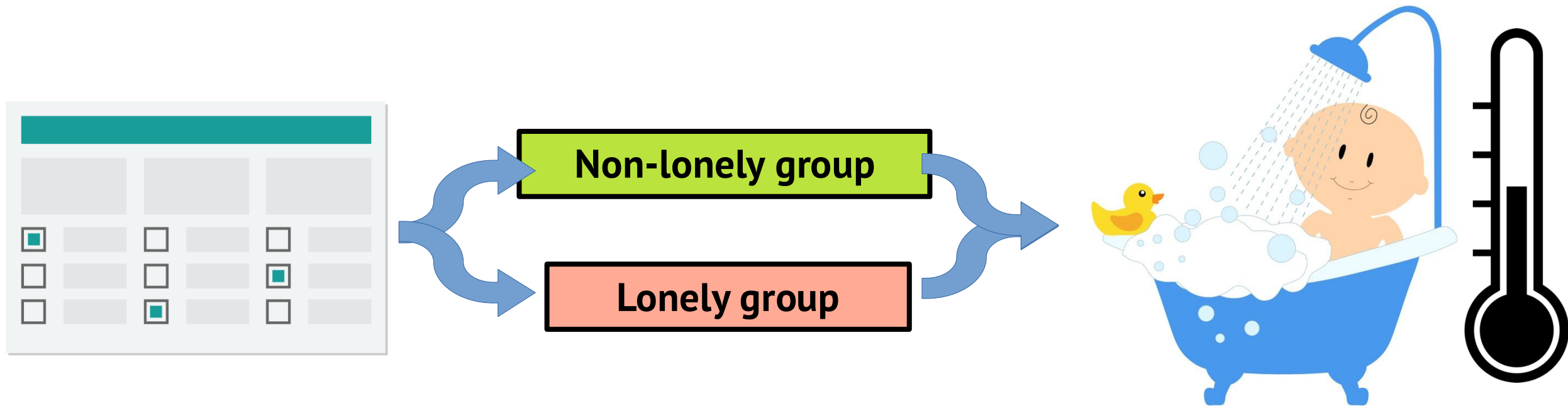
Lonely group

“... being an introvert is not really a good thing for relationships.”

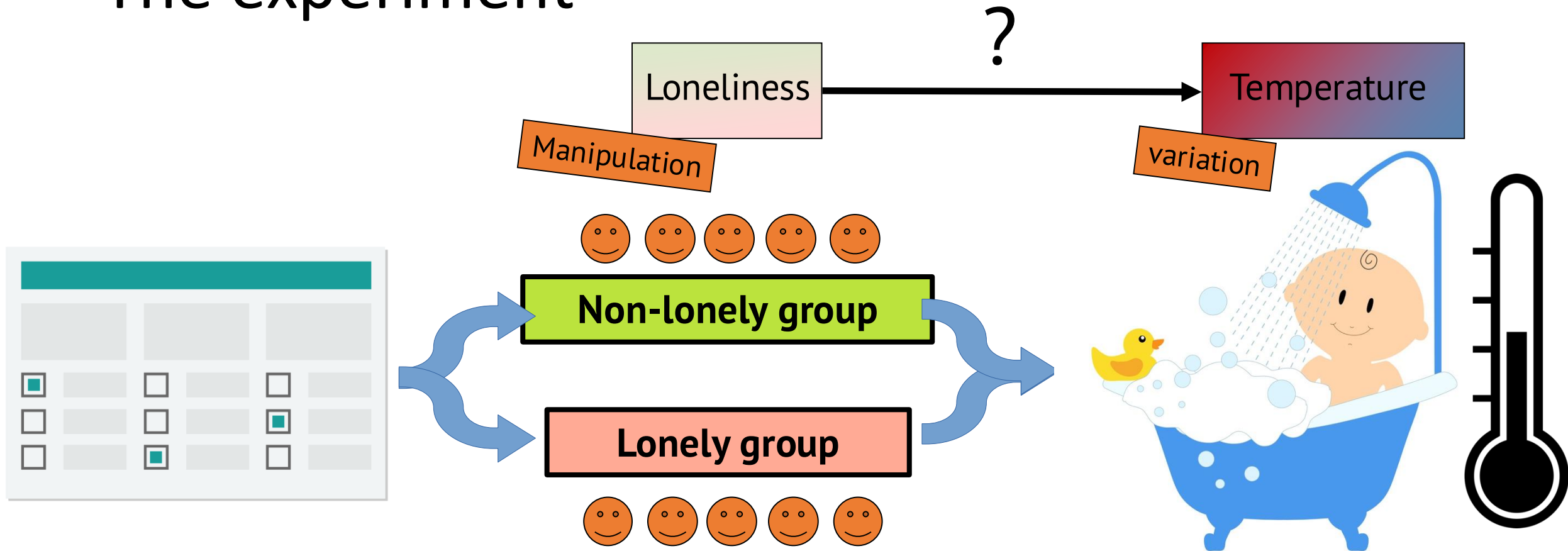
Non-lonely group

“... being an introvert is a really good thing for relationships.”

The experiment



The experiment



Question: What type of variables?

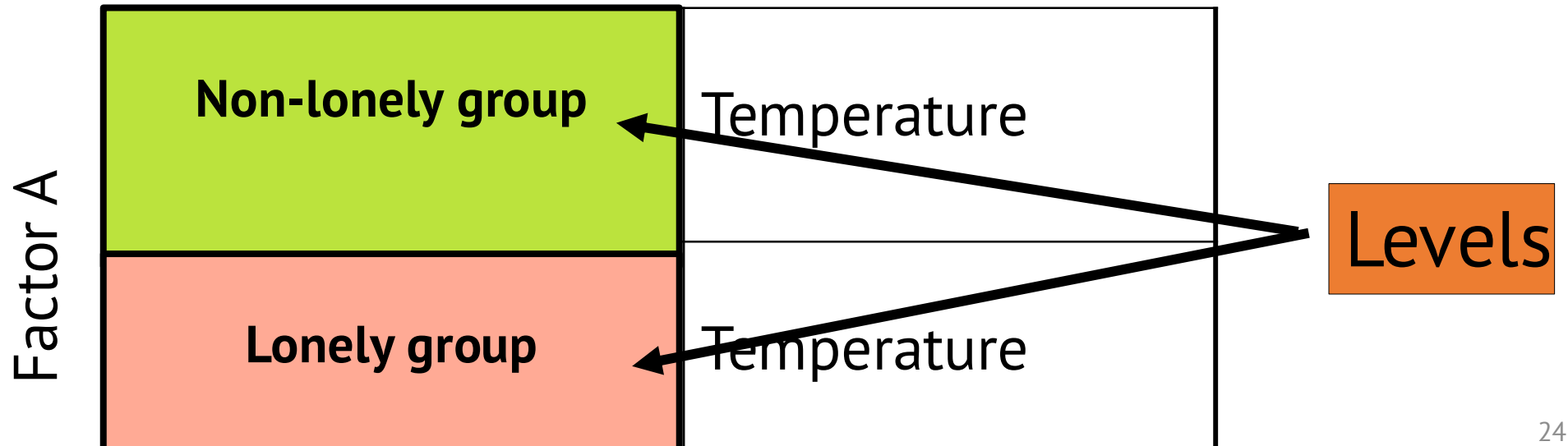
- Loneliness
 - Independent variable (IV, the book also uses “manipulated variable”)
 - Categorical (because two groups)
- Temperature
 - Dependent variable (DV, the book also uses “measured variable”)
 - Quantitative

		IV	
		Quantitative	Categorical
DV	Quantitative	Correlation Regression	t-test ANOVA
	Categorical	Logistic regression	Contingency table

One-way experimental design



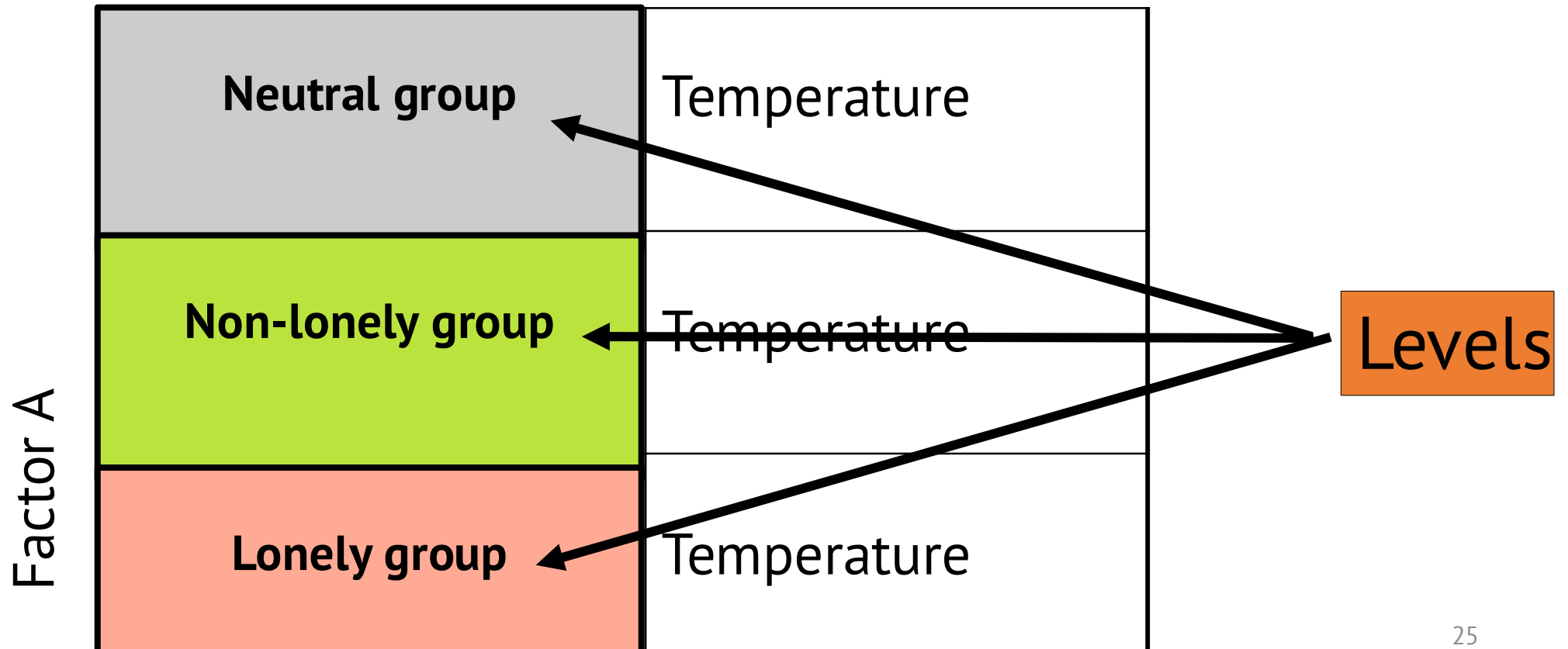
Design:



One-way experimental design

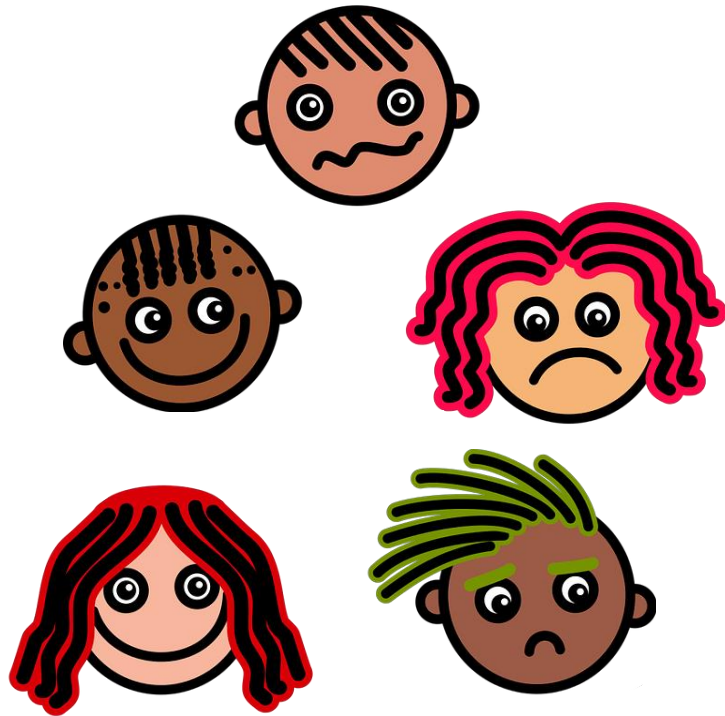


Design:

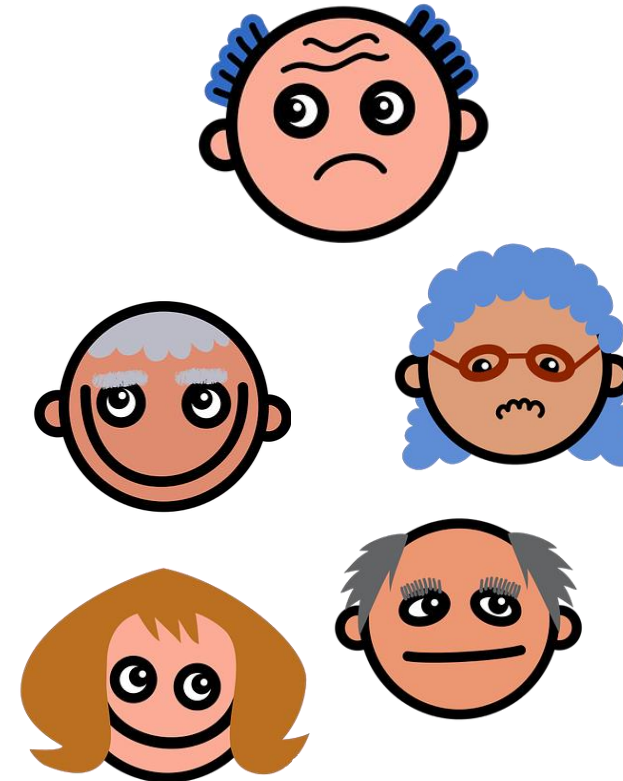


Collecting participants and assigning conditions

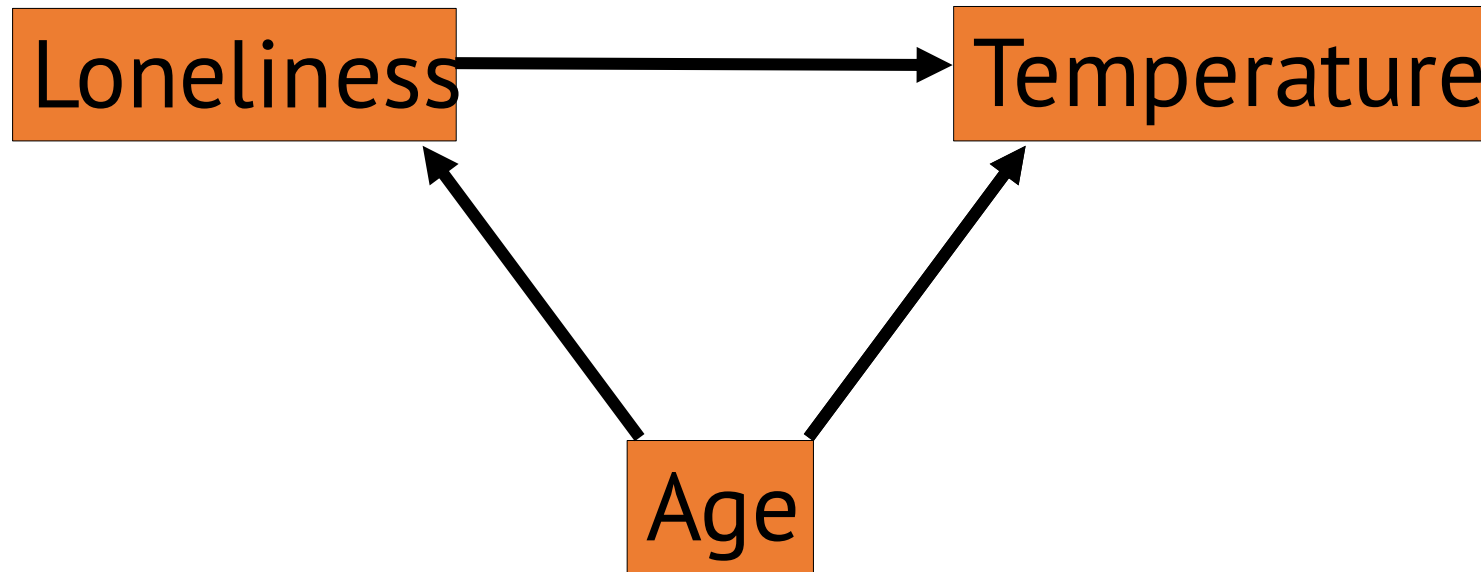
Non-lonely group



Lonely group



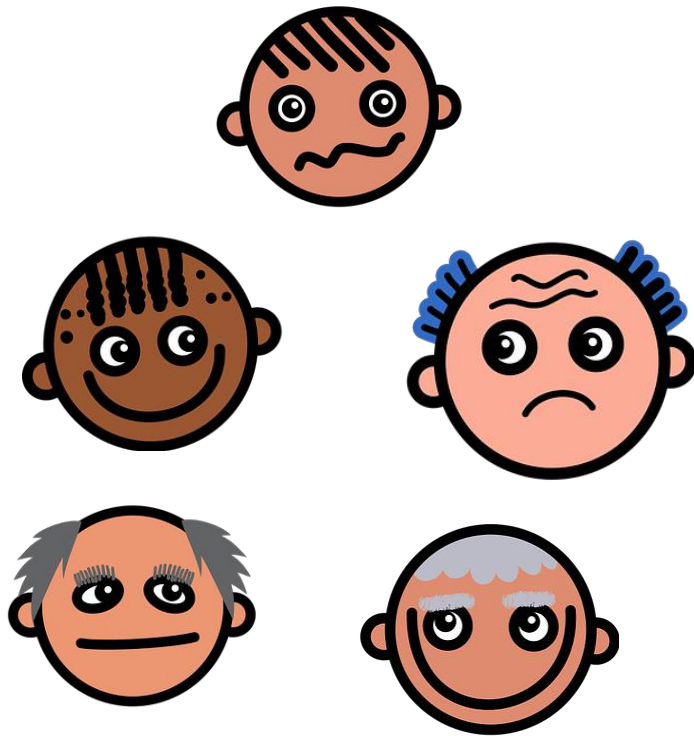
Systematic difference between group: potential confound



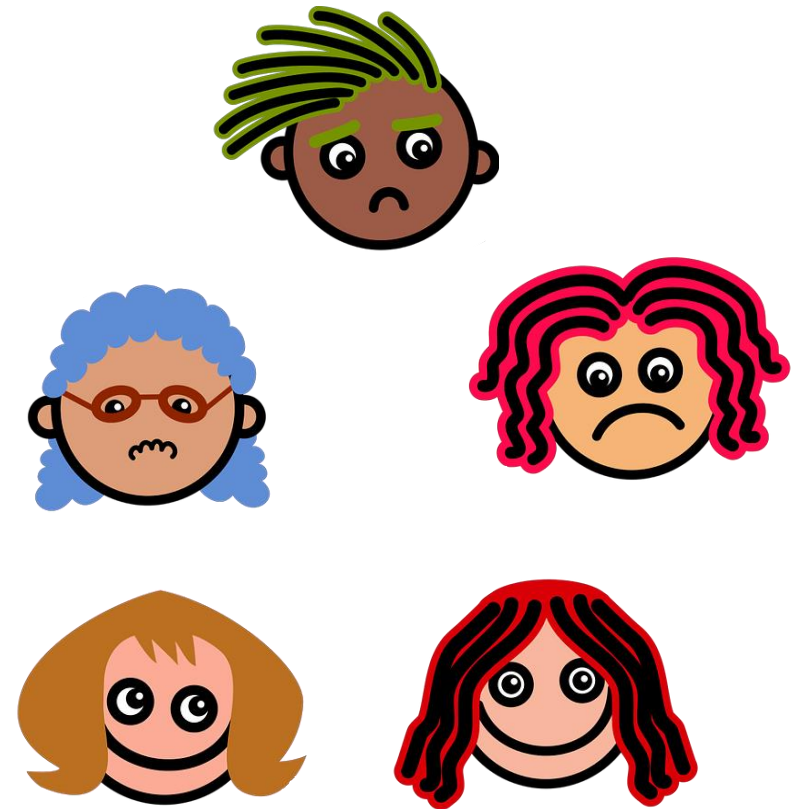
Internal validity: Are there alternative explanations for the statistical effect?

Collecting participants and assigning conditions

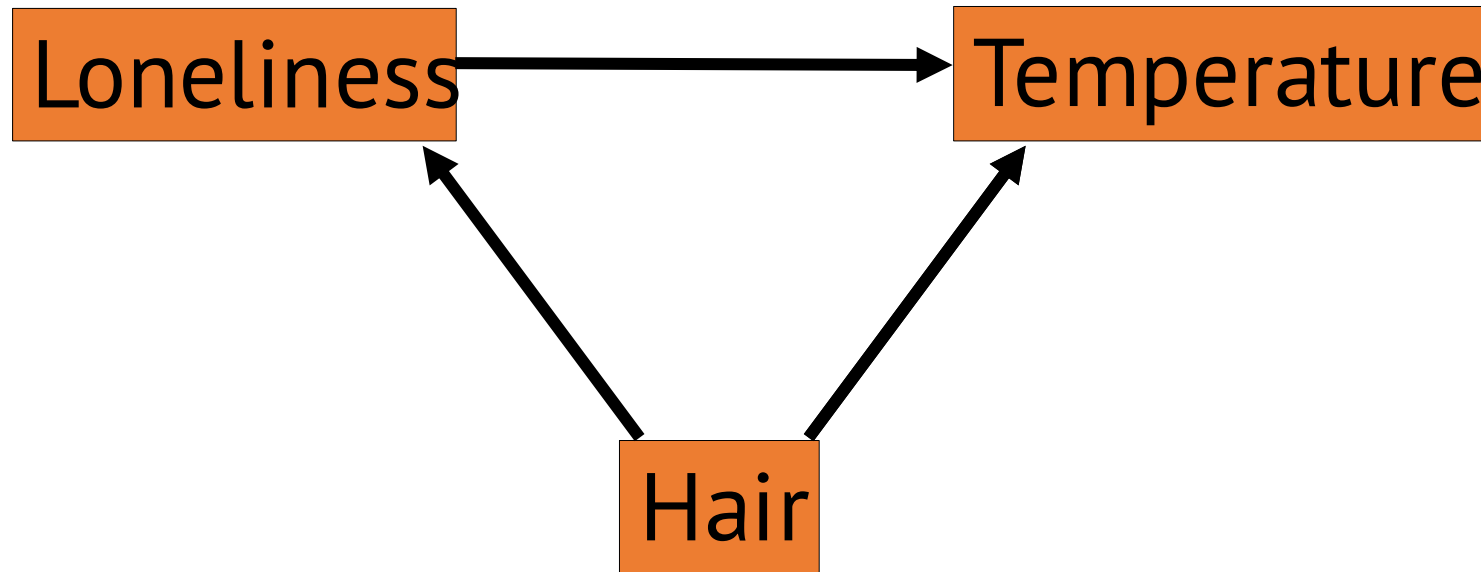
Non-lonely group



Lonely group



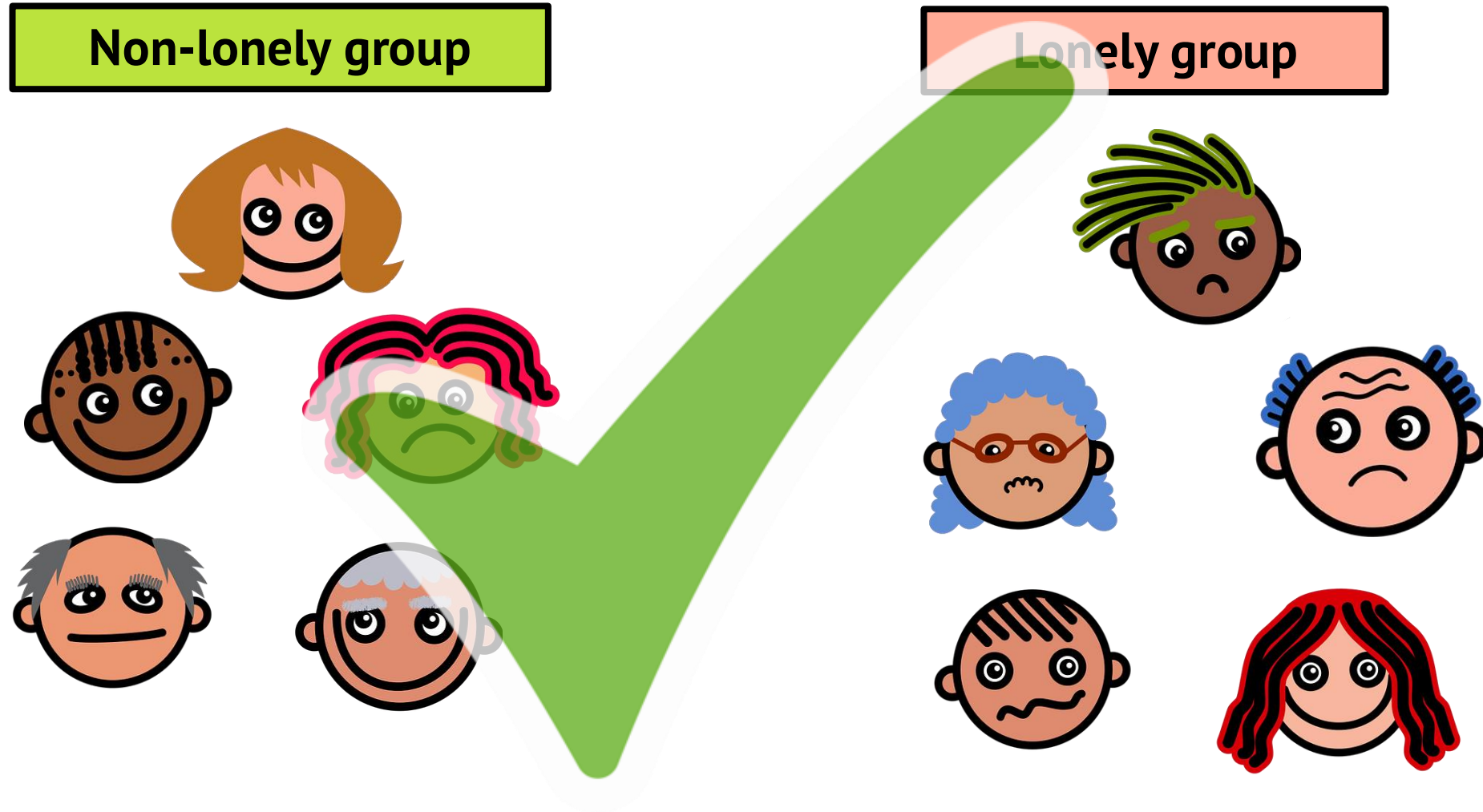
Systematic difference between group:
potential confound



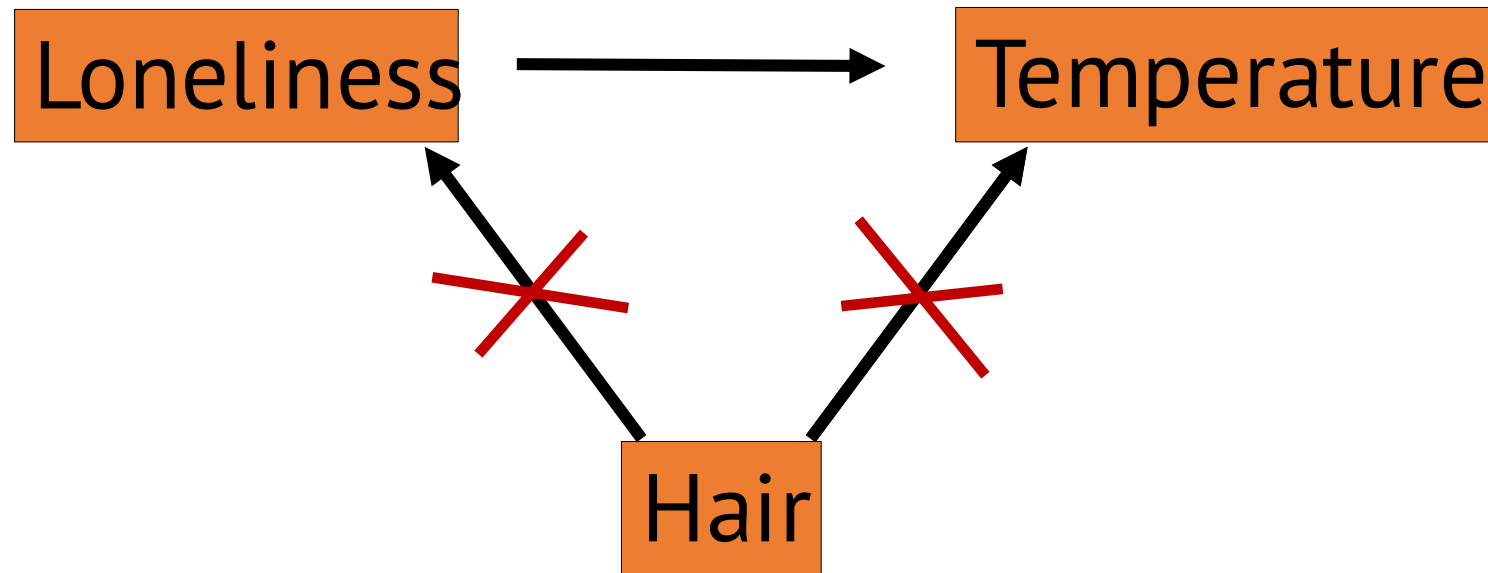
Solution: Experimental groups

- Balanced groups: Only systematic difference between the group is the manipulation
- Method: **Random assignment**
 - Expected value of the difference between groups for all background variables should be roughly 0:
 - Age $\approx$ age
 - Hair $\approx$ hair

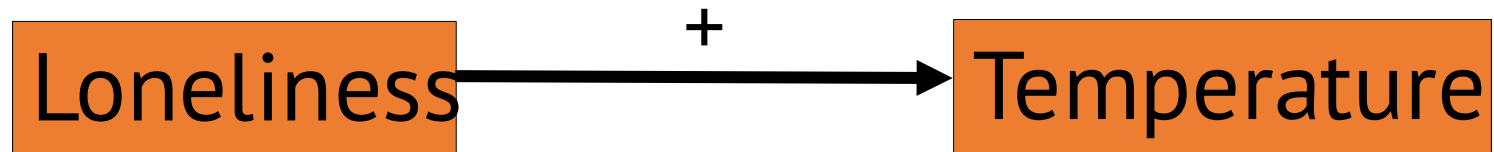
Collecting participants and assigning conditions



Random assignment precludes confounds



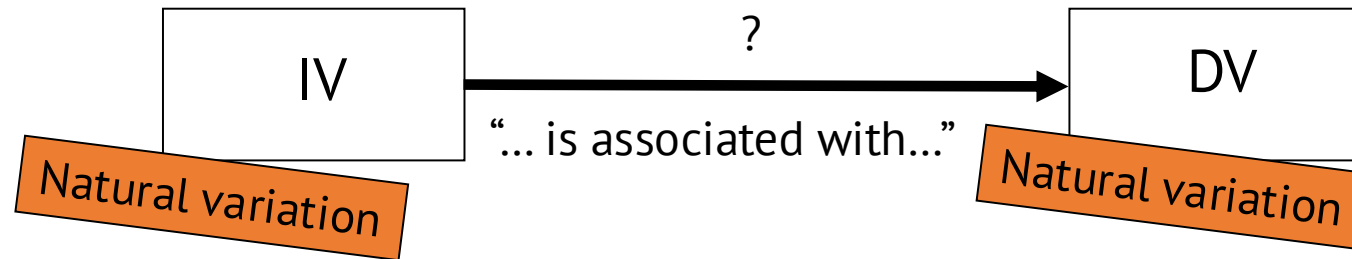
Temporal precedence establishes direction



- FIRST we manipulated loneliness
- THEN we measured water temperature

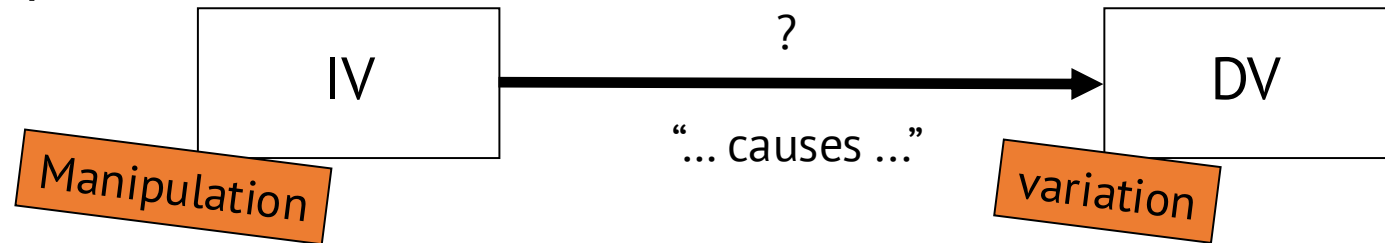
Differences between observational and experimental research

Observational research



Experimental research enables causal claims!

Experimental research



Differences between observational and experimental research

Cons

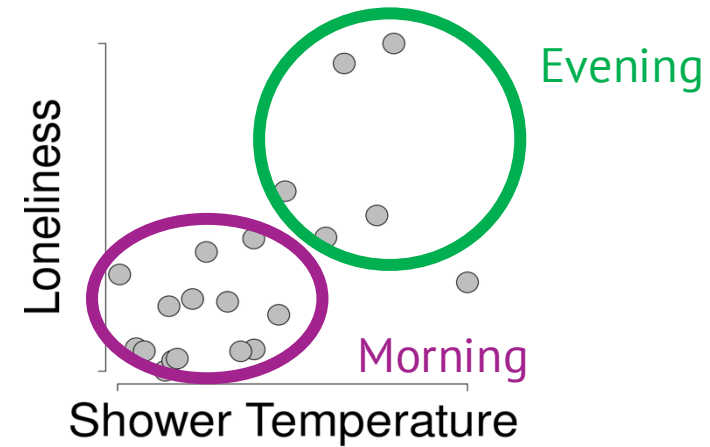
Experimental	Observational
* Impossible	
* Impractical	
* Unethical	
* Unnatural	



Differences between observational and experimental research

Cons

Experimental	Observational
* Impossible	* Confounds
* Impractical	* No causality
* Unethical	
* Unnatural	



Differences between observational and experimental research

Cons

Experimental	Observational
* Impossible	* Confounds
* Impractical	* No causality
* Unethical	
* Unnatural	

Pros

Experimental	Observational
* Control	* Easier to do
* Causality	* Natural variation

*Generalizability: When a theory is supported by both observational and experimental studies
→ Replication!*

Today

1. Experimental studies vs observational studies
2. **Pitfalls in experimental studies**
 1. Pseudoreplication
3. Different experimental designs
 1. Pretest
 2. Within groups vs between groups
4. Order effects
 1. Counterbalancing
5. Recap
 1. Next time
 2. Example exam question

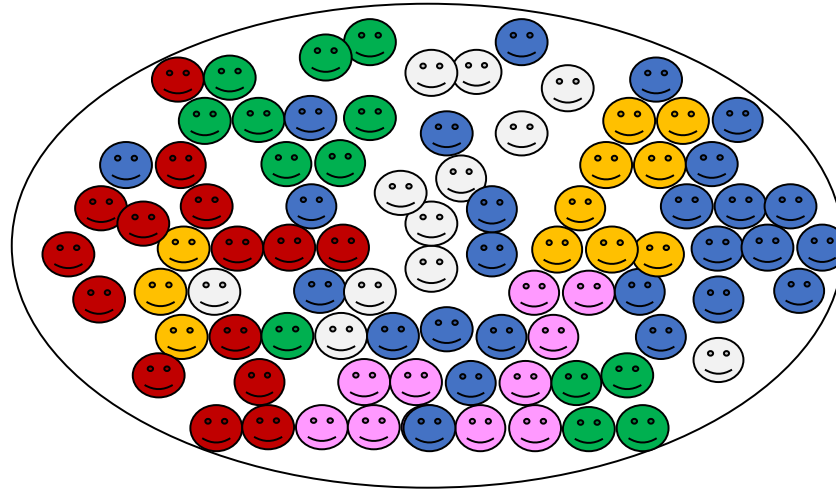
Pitfalls of experimental design

1. Manipulation changes multiple variables
 - Design confound
2. There are other systematic differences between groups
 - Non-random assignment
 - Participants can choose which condition
 - First six in the Lonely group, last six in the Non-lonely group
 - First six in the morning, last six in the evening
 - ...
 - Selection effect → Confound

Pitfalls of experimental design

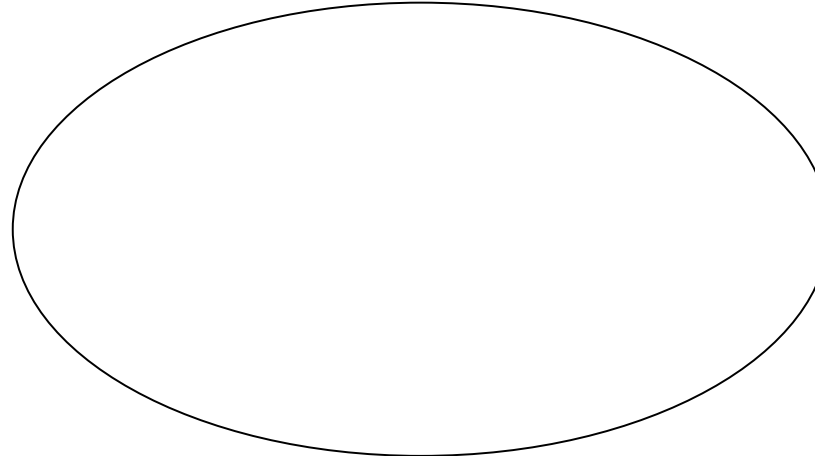
1. Manipulation changes multiple variables
 - Design confound
2. There are other systematic differences between groups
 - Non-random assignment
 - Selection effect → Confound
3. Pseudoreplication
 - Amplifies confounds

Population

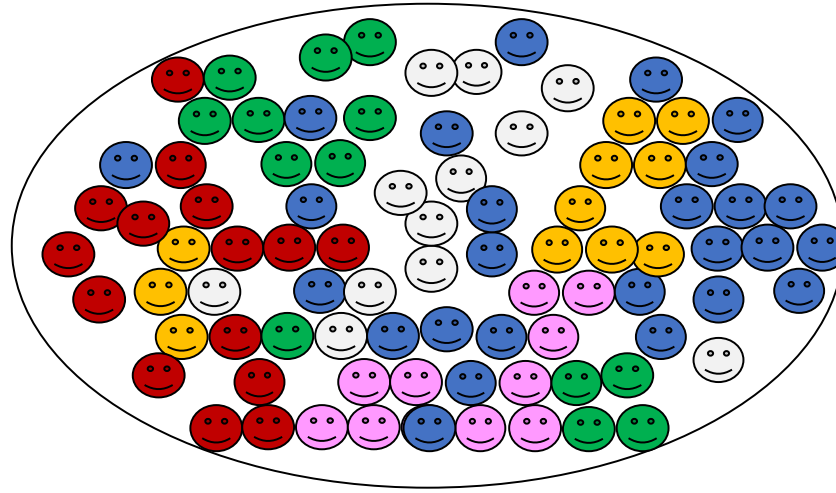


Replication: all observations are *independent*

Sample

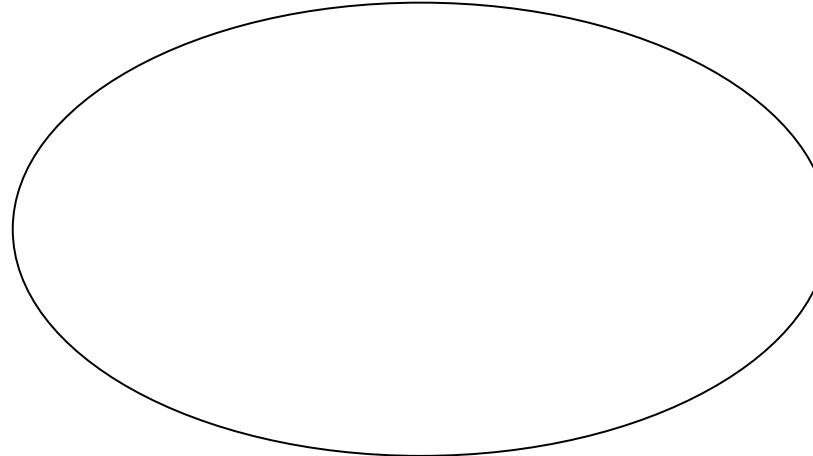


Population



Pseudoreplication: the observations are *dependent*

Sample



What causes pseudoreplication?

- Observations are from the same environment
 - Children in one class
 - Students from one bachelor
 - Plants in one greenhouse
 - Trees in one forest
- Observations are genetically related
 - Siblings
 - Pups from one rat genetic strain
- Observations are related in time
 - Repeated measures (analyzed as if independent)

Pseudoreplication

Non-lonely group



Lonely group

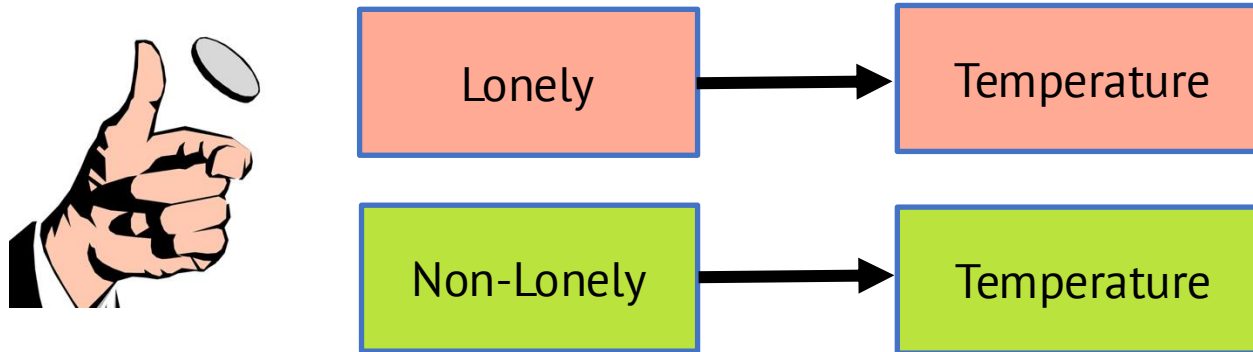


Today

1. Experimental studies vs observational studies
2. Pitfalls in experimental studies
 1. Pseudoreplication
3. **Different experimental designs**
 1. Pretest
 2. Within groups vs between groups
4. Order effects
 1. Counterbalancing
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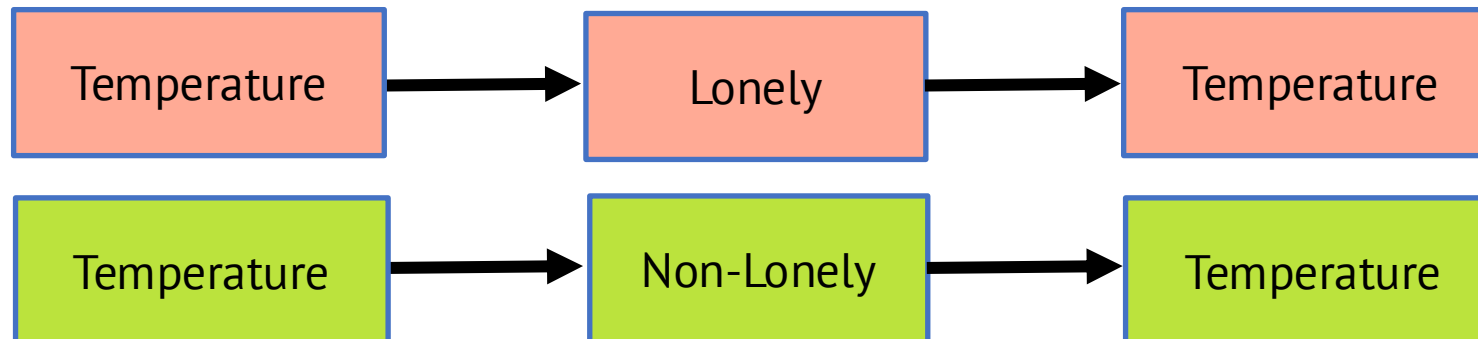
Different experimental designs

- So far: Posttest-only design
 - Random assignment in multiple groups
 - **Assume** no difference in DV prior to group assignment
 - Test the difference in DV after group assignment



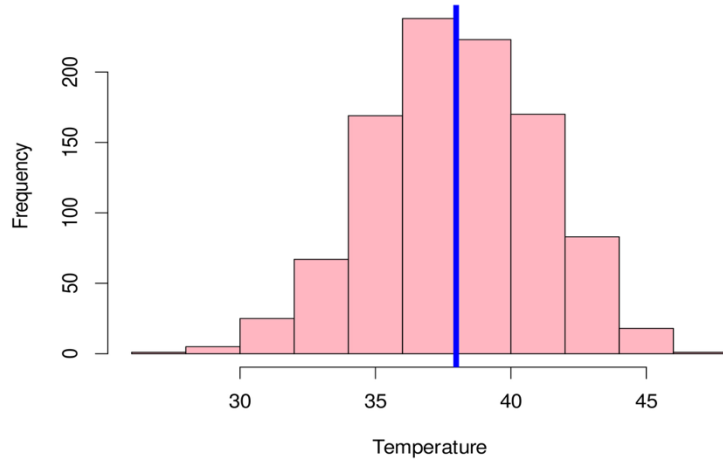
Different experimental designs

- Pretest/Posttest design
 - Random assignment in multiple groups
 - Measure DV before the manipulation
 - Measure DV after the manipulation
 - Test the “difference of the difference”

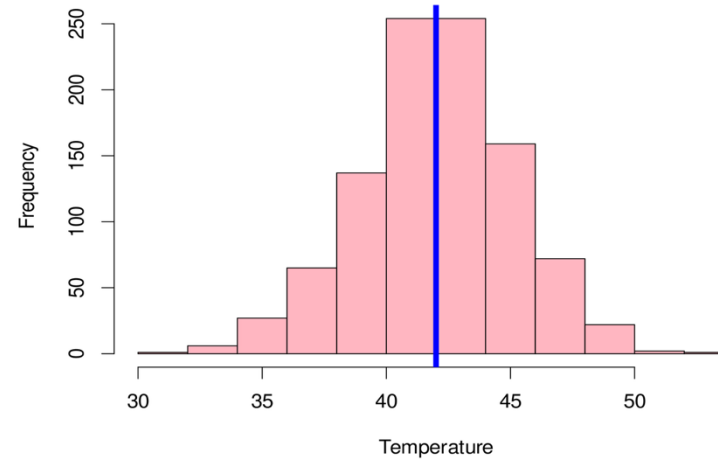


Pretest/Posttest Design

Pre-Test Lonely: mean = 37.98

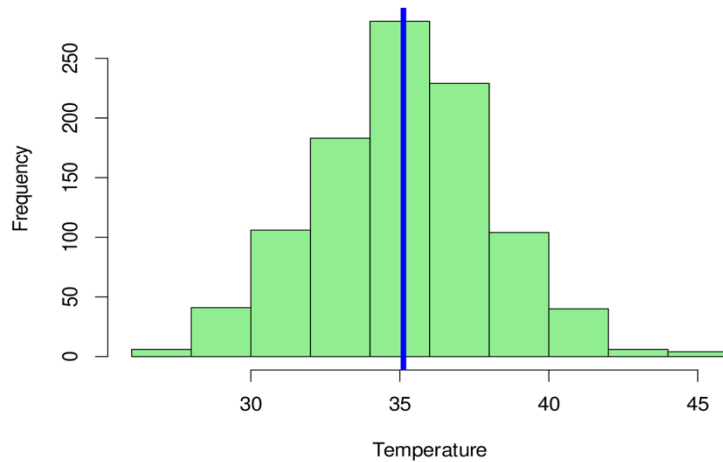


Post-Test Lonely: mean = 42

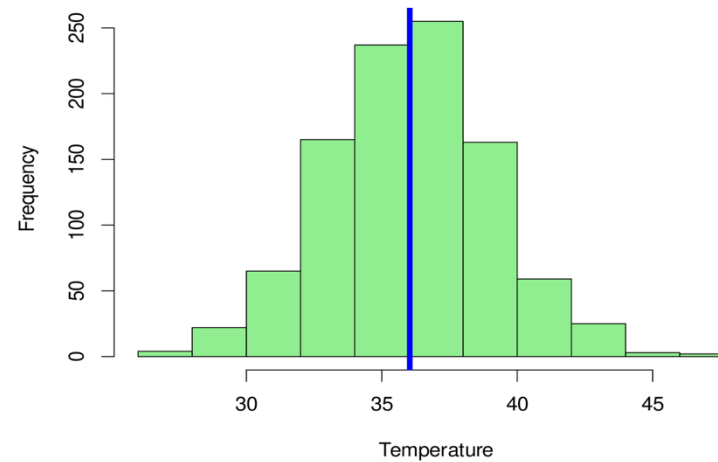


$$\text{Difference: } 42 - 37.98 = 4.02$$

Pre-Test Non-Lonely: mean = 35.12



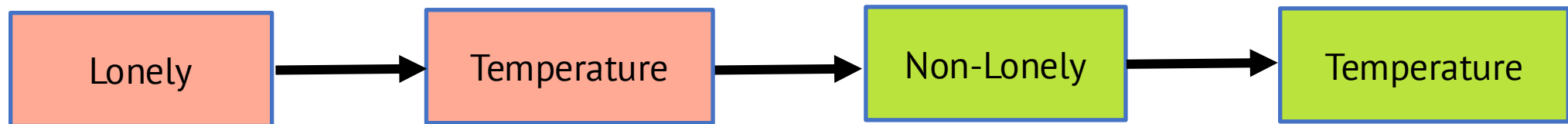
Post-Test Non-Lonely: mean = 36.03



$$\text{Difference: } 36.03 - 35.12 = 0.91$$

Different experimental designs

- Repeated-measures design
 - One of two within-groups designs (the other: concurrent-measures)
 - Measure DV **twice in each participant**
 - Test the difference in DV



Between groups design

Non-lonely group

Lonely group

Differences
due to background
variables (e.g., age,
freq. sauna visits)

Carla	40°C
Henk	37°C
Miep	43°C
Joost	40°C
Tim	37°C
Fleur	43°C
Mean:	40°C

39°C Piet
41°C Bep
43°C Floor
41°C Gerda
45°C Hans
42°C Ron
Mean: 42°C

Differences
due to background
variables (e.g., age,
freq. sauna visits)

Difference in mean driven by background variables AND manipulation

Matched groups design

(Morling, pp. 291, 297)

- Participants who are similar on some measured variable are grouped into sets; the members of each set are then randomly assigned to different conditions
- E.g., pairs of similar age, randomly split over Lonely and Non-lonely
- Use: ensures the groups are equal on age before the manipulation
- Still different people per group, but can be treated as a within-groups design
- Requires more time and often more resources than random assignment

Within groups design

Non-lonely group

Lonely group

Differences
due to background
variables (e.g., age,
freq. sauna visits)

But not between groups

Carla 40°C	39°C Carla
Henk 37°C	41°C Henk
Miep 43°C	43°C Miep
Joost 40°C	41°C Joost
Tim 37°C	45°C Tim
Fleur 43°C	42°C Fleur
Mean: 40°C	Mean: 42°C

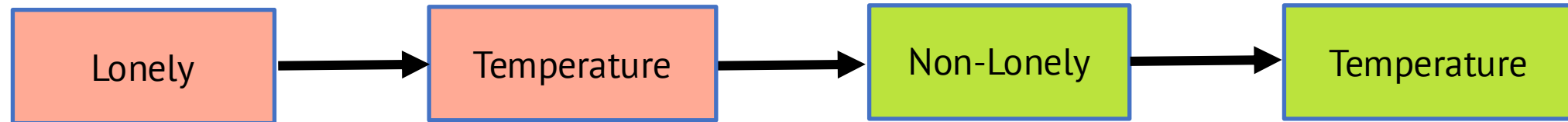
Difference in mean driven by ~~background variables~~ AND manipulation

Today

1. Experimental studies vs observational studies
2. Pitfalls in experimental studies
 1. Pseudoreplication
3. Different experimental designs
 1. Pretest
 2. Within groups vs between groups
4. **Order effects**
 1. Counterbalancing
5. Recap
 1. Next time
 2. Example exam question

Order effects

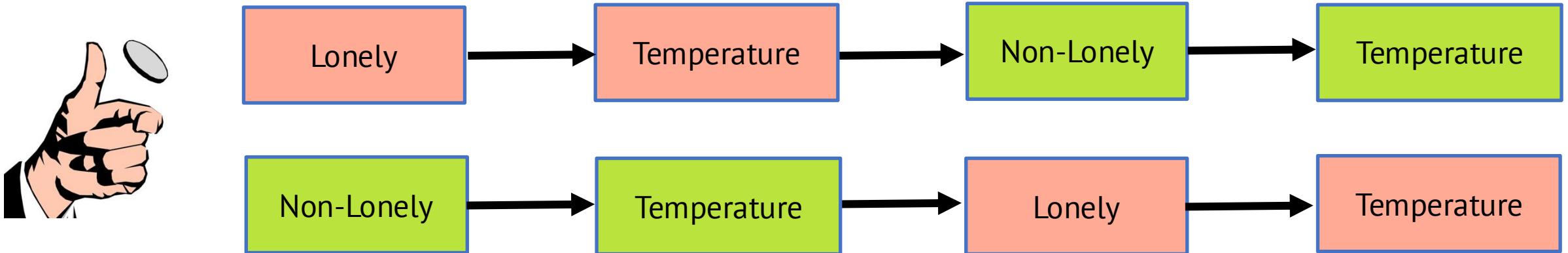
- Repeated-measures design



- Problems:
 - Carryover effects
 - Practice (fatigue) effects
 - ...

Solution: counterbalancing

- Repeated-measures design



- Order effects remain for each individual, but *not* the group

Four validities for our experiment

(Morling, p. 303)

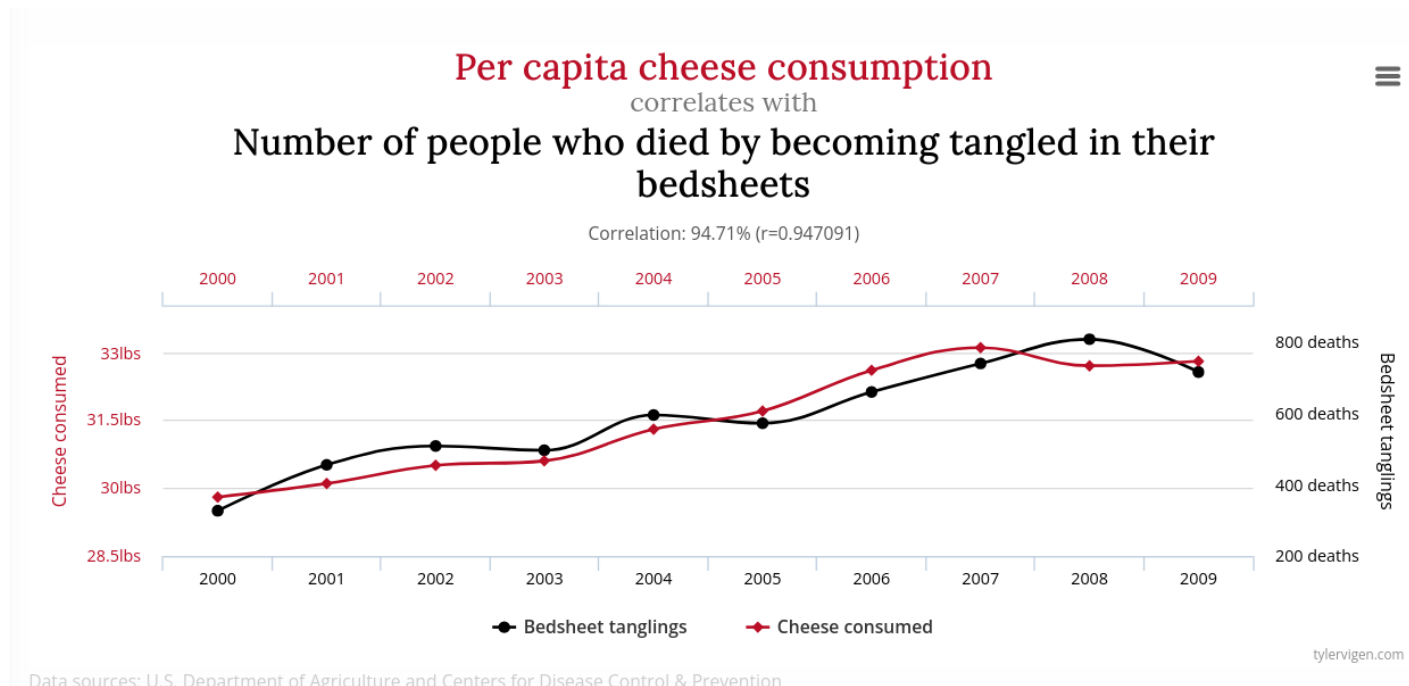
- **Construct validity**: how well were the variables measured and manipulated?
 - Did the feedback really make people lonelier? → manipulation check
- **External validity**: to whom or what can the causal claim generalize?
 - Other people, places, and times? E.g., everyday showers at home
- **Statistical validity**: how much? How precise? What else is known?
 - How big is the temperature difference? Has it been replicated?
 - Bargh & Shalev's (2012) Study 1 failed to replicate (Donnellan et al., 2015)
- **Internal validity**: are there alternative explanations for the results?
 - Design confounds? Selection effects? Order effects?

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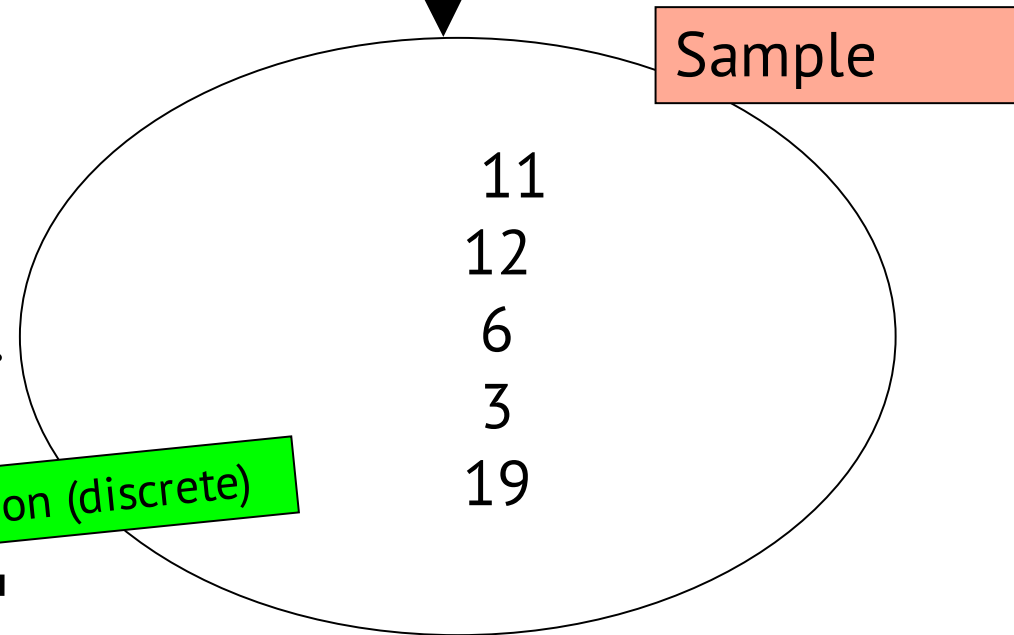
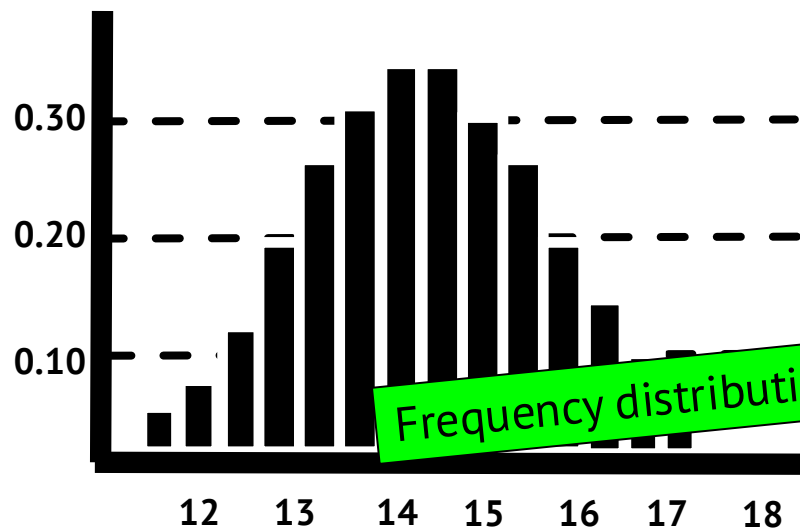
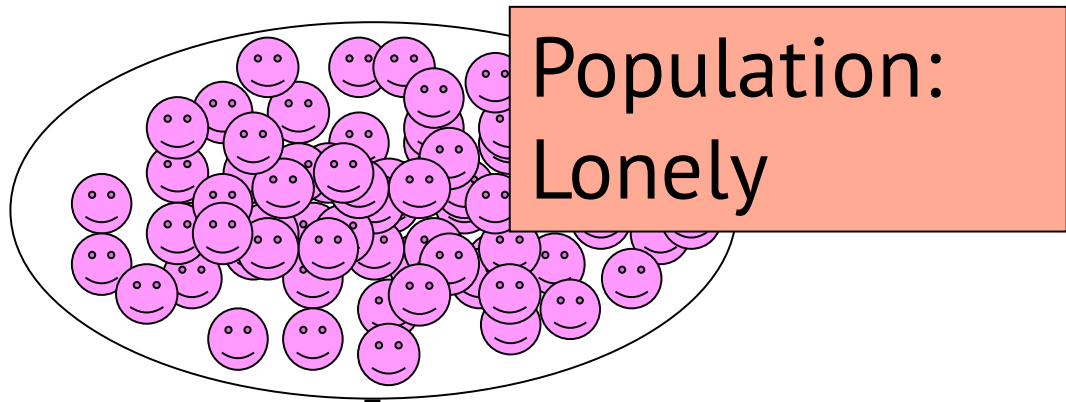
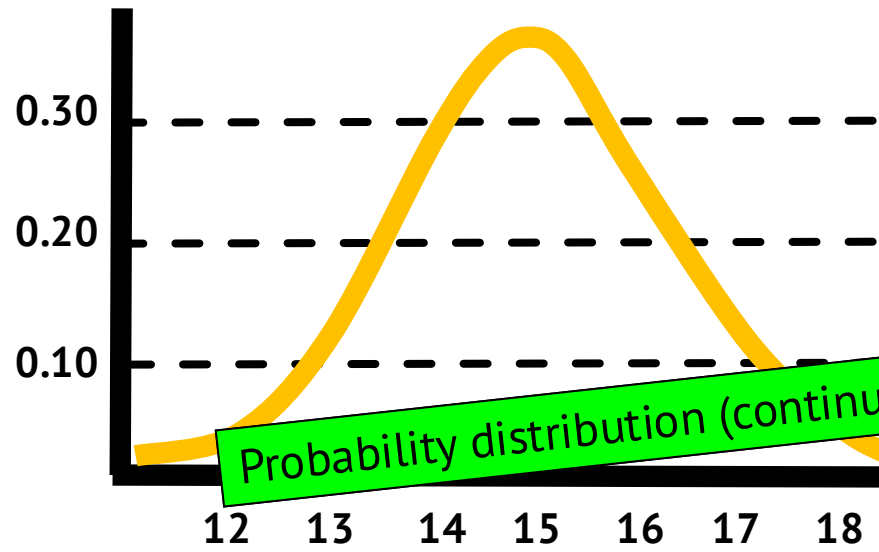
Your Research Questions

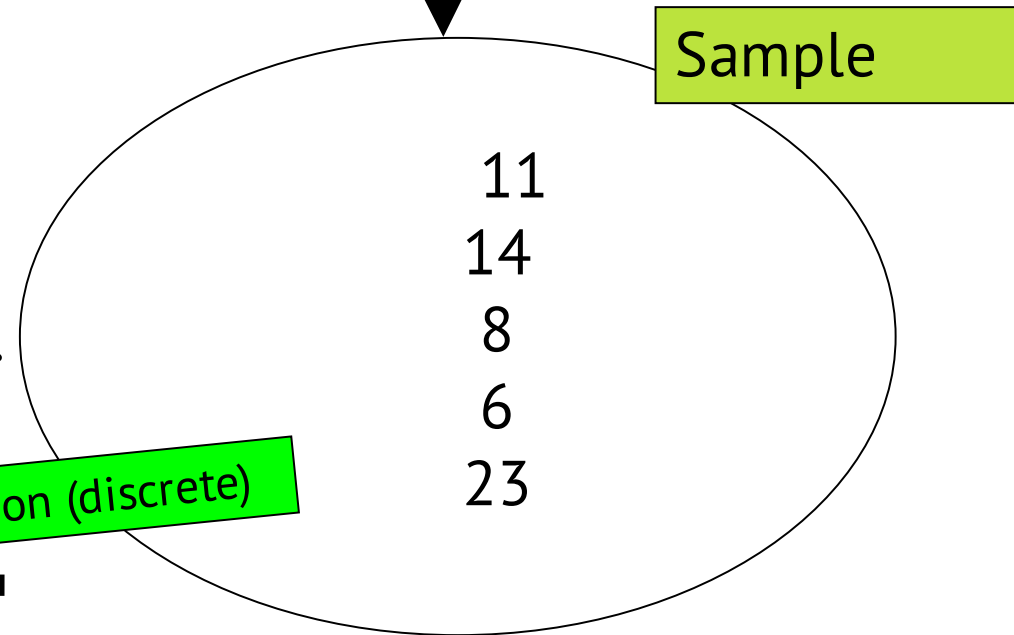
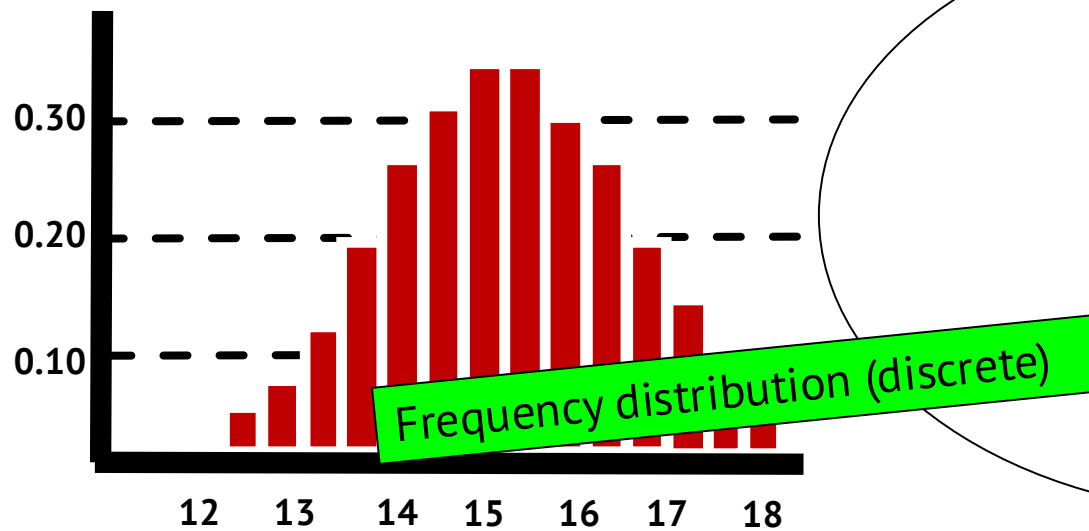
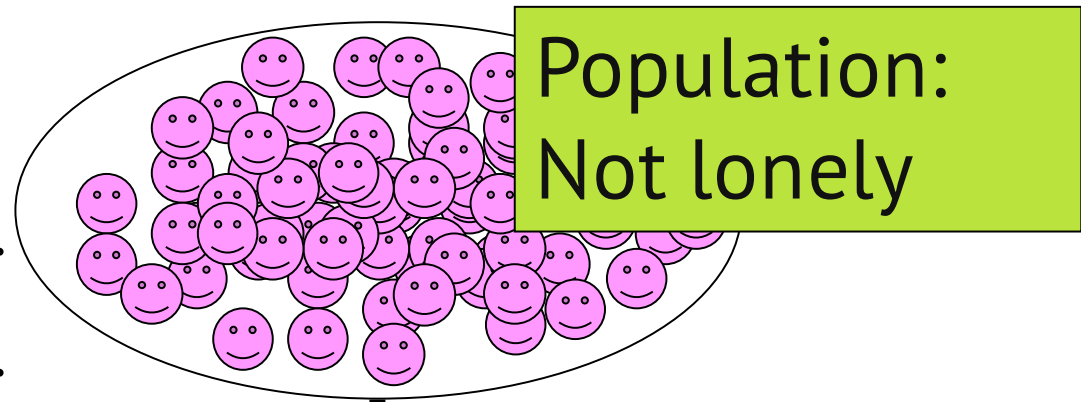
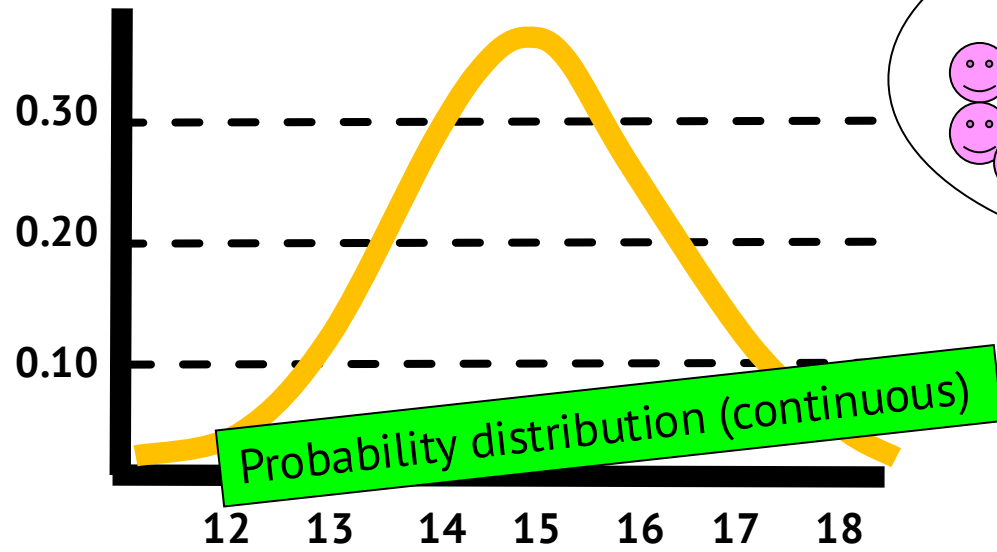
- Think of two (psychological) variables you would like to investigate
- How would you test a **causal** association?



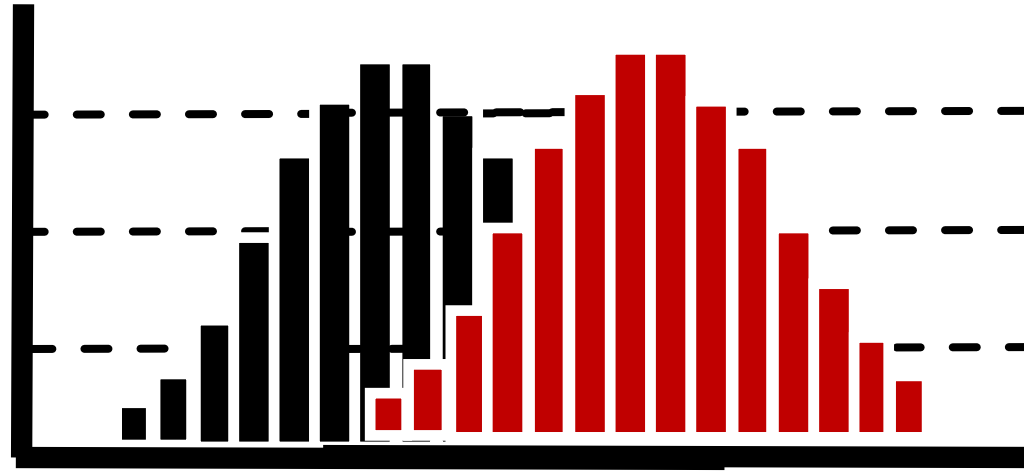
Recap

- In order to make causal claims, we can do experimental studies
- Watch out for:
 - Design confounds → experimental control
 - Selection effects → random assignment
 - Different baseline levels → pretest/posttest
 - Different backgrounds → repeated measures





Next week



- → What evidence do we have that there is a difference between these two populations?

Next week



Inferential Statistics!!

- → We will see the difference between

Today

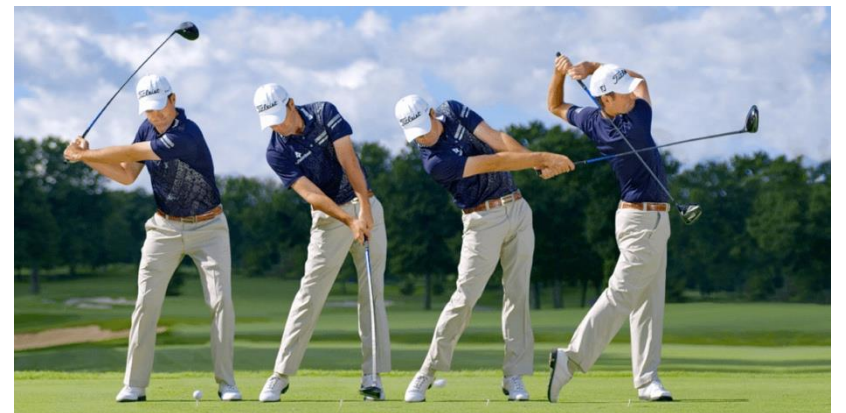
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 2. **Example exam question**

Example exam question

Imagine a researcher who is interested in the effect of meditation on people's ability to improve their golf swing. She sets up an experiment in which 30 novice golfers are tested. First, all of them are asked to drive a golf ball as far as possible (5 times). Next, 15 randomly chosen golfers are asked to meditate for 10 minutes, while the other 15 are given a newspaper to read. After 10 minutes, all golfers are again asked to drive a golf ball as far as possible (5 times).

What is the experimental design described here?

- a) Posttest-only design
- b) Pretest/posttest design
- c) Repeated measures design

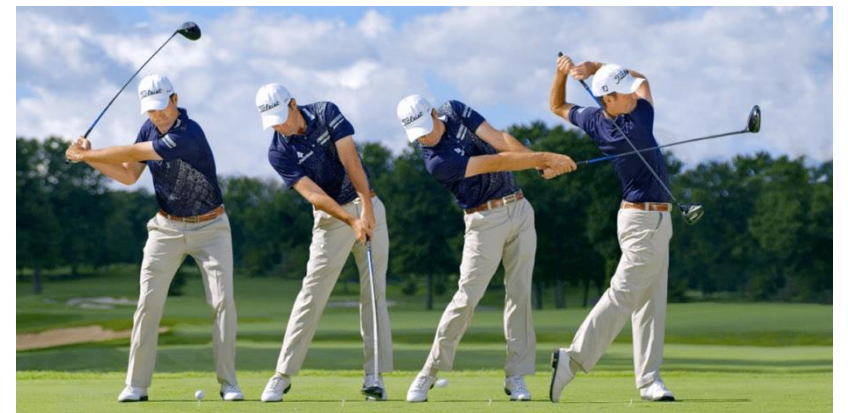


Example exam question

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What is the experimental design described here?

- a) ~~Posttest-only design~~
- b) Pretest/posttest design
- c) ~~Repeated measures design~~



Questions?

Thank you for your attention

